

Understanding Density Transformation

The density transformation formula you're looking at is:

$$\rho(x) = J_y(x)\mu(y(x))$$

Where:

- $\rho(x)$ is the density of the original variable x .
- $\mu(y(x))$ is the density of the transformed variable y .
- $J_y(x)$ is the Jacobian determinant of the transformation $y(x)$.

Understanding the Components

1. Density of the Original Variable $\rho(x)$

This represents how likely you are to find a particular value of x in your original data distribution.

2. Density of the Transformed Variable $\mu(y(x))$

This is the density function for the new variable y , which is defined such that it may have properties (like being Gaussian) that are easier to work with or known.

3. Jacobian $J_y(x)$

The Jacobian measures how the transformation affects the “volume” in the space of x compared to the space of y . Essentially, it tells you how much the area (or volume in higher dimensions) expands or contracts when transforming from x to y .

Why the Jacobian Matters

When you transform variables, the probability densities can change because you're essentially changing how space is “filled” by those values. The Jacobian adjusts for this change in density.

Example 1: Simple Linear Transformation

Transformation: Suppose you have a simple linear transformation defined by $y = 2x$.

- **Original density:** Let's say x is uniformly distributed between 0 and 1, so $\rho(x) = 1$ for $0 \leq x < 1$.
- **New variable:** For $y = 2x$, the new variable y will range from 0 to 2.
- **Density of y :** The transformation stretches the space. For any given y value, there are fewer x values that will map to it.
- **Jacobian:** The derivative of the transformation $y = 2x$ gives the Jacobian:

$$J_y(x) = \frac{dy}{dx} = 2$$

- **Calculating density:**

$$\mu(y) = (\text{for } 0 \leq y < 2) \quad \mu(y) = \frac{1}{2} \quad (\text{since it's uniform over } [0,2])$$

Using the formula:

$$\rho(x) = J_y(x)\mu(y(x)) = 2 \cdot \frac{1}{2} = 1$$

This confirms that the density of x remains consistent with its uniform distribution.

Why Volume Correction?

When transforming from x to y , the distribution of points can become less dense (or more dense) depending on how the transformation stretches or compresses the space. The Jacobian helps to correct for this by scaling the density appropriately.

Change of Variables Formula

The change of variables formula for probability densities is given by:

$$\rho(x) = \mu(y(x)) \cdot |\det(Dy(x))|$$

Explanation of Terms

- $\rho(x)$: This is the probability density function (PDF) of the variable X at point x .
- $\mu(y)$: This is the probability density function of the variable Y at point y , where $y = y(x)$ is the transformed version of x through the function $y(x)$.
- $y(x)$: This represents the transformation function that maps x to y . It could be any continuous and differentiable function.
- $Dy(x)$: This is the Jacobian matrix of the transformation $y(x)$ at point x . It contains all the first-order partial derivatives of the function and describes how the function changes as you vary x .
- $|\det(Dy(x))|$: This is the absolute value of the determinant of the Jacobian matrix. It quantifies how the transformation $y(x)$ changes volumes (or densities) in the space. The determinant indicates how the area (in 2D) or volume (in higher dimensions) scales under the transformation.

Relationship to Density Transformation

The change of variables formula is fundamentally about how probabilities (or densities) transform under a change of variables:

- **Density Transformation:** When you transform a random variable from one space to another using a function, the density must be adjusted to account for how the volume (or area) changes due to the transformation. The Jacobian determinant $|\det(Dy(x))|$ acts as a scaling factor that corrects the density accordingly.
- **From High Dimensions to Latent Space:** In generative modeling, such as with normalizing flows or models like NICE, you often transform data from a complex high-dimensional space to a simpler latent space. The change of variables formula allows you to compute the density of the data in the original space based on the known density in the latent space.

Example

To illustrate, suppose you have a one-dimensional transformation defined as $y(x) = f(x)$:

1. If $\mu(y)$ is known for some values of y , you can compute $\rho(x)$ using the change of variables formula.
2. The Jacobian would be $Dy(x) = f'(x)$, and thus $|\det(Dy(x))| = |f'(x)|$.
3. Plugging into the formula gives:

$$\rho(x) = \mu(f(x)) \cdot |f'(x)|$$